

INDIAN STATISTICAL INSTITUTE

Semester Examination 2026

M.Math Second year, Second Semester

Symplectic Geometry

Maximum Marks: 60

Date: 2 May, 2026

Duration: 3 hours

Notations: $SO(n)$ is the special orthogonal group
 G is a Lie group and $\mathfrak{g}$ its Lie algebra

- (1) Let M be a smooth manifold and T^*M its cotangent bundle. Use the tautological 1-form on T^*M to define a contact form on $T^*M \times \mathbb{R}$. Prove that there is an infinite family of Legendrian submanifolds. 10

- (2) Consider the 1-form $\alpha = \cos r \, dz + r \sin r \, d\varphi$ on $\mathbb{R}^3$, where (r, φ, z) denotes the cylindrical coordinate system on $\mathbb{R}^3$.

(a) Prove that α is a contact form on $\mathbb{R}^3$.

(b) Find a 2-parameter subgroup of diffeomorphisms which preserve the associated contact structure. 10

- (3) (a) Let D be a distribution on $\mathbb{R}^3$ defined by the 1-form $\alpha = dz - y \, dx$. Find vector fields X, Y in D such that $X(x), Y(x)$ span D_x for all $x \in \mathbb{R}^3$. Show that the vectors $X(x), Y(x), [X, Y](x)$ is a basis of $\mathbb{R}^3$ for all x .

(b) Suppose M is a smooth manifold of dimension 3 admitting vector fields X, Y such that $X(x), Y(x), [X, Y](x)$ is a basis of $T_x M$ for all $x \in M$. Prove that M is a contact manifold. 10

- (4) Let $\lambda_1, \lambda_2 > \lambda_1$ be two real numbers. Consider the function $\rho : \mathbb{CP}^1 \rightarrow \mathbb{R}$ defined by

$$\rho([z_1 : z_2]) = \frac{\lambda_1 |z_1|^2 + \lambda_2 |z_2|^2}{|z_1|^2 + |z_2|^2}, \quad [z_1, z_2] \in \mathbb{CP}^1.$$

Is the 2-form $\omega = \frac{i}{2} \partial \bar{\partial} \rho$ on $\mathbb{CP}^1$ symplectic? Justify your answer. 12

- (5) Let G act on two symplectic manifolds (M_1, ω_1) and (M_2, ω_2) . Suppose the actions are Hamiltonian, and $\mu_j : M_j \rightarrow \mathfrak{g}^*$, $j = 1, 2$, are the moment maps of the actions.

Prove that the diagonal action of G on $M_1 \times M_2$ is Hamiltonian and its moment map μ is given by

$$\mu(p_1, p_2) = \mu_1(p_1) + \mu_2(p_2) \quad \text{for } (p_1, p_2) \in M_1 \times M_2.$$

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(6) (a) Prove that the tangent space of $SO(3)$ is the space of all 3×3 skew-symmetric matrices $\mathfrak{o}(3)$. Prove that $\mathfrak{o}(3)$ is a Lie subalgebra of $M_3(\mathbb{R})$.

(b) Show that $\mathbb{R}^3$ has a Lie algebra structure given by $[a, b] := a \times b$ for all $a, b \in \mathbb{R}^3$. Denote it by $\mathfrak{h}$. Prove that $\phi : \mathfrak{h} \rightarrow \mathfrak{o}(3)$ defined by

$$a = (a_1, a_2, a_3) \mapsto A = \begin{pmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{pmatrix}$$

is a Lie algebra isomorphism.

(c) Consider the induced adjoint action of $SO(3)$ on $\mathfrak{h}$ and $\mathfrak{h}^*$. Show that the orbits are symplectic manifolds.

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$$[A, [B, C]] + [B, [C, A]] + [C, [A, B]]$$

$$\cancel{ABC} \rightarrow \cancel{CBA}$$

$$AB/C - A/CB - B/CA + C/BA$$

$$+ B/CA - B/AC - CA/B + CA/CB$$

$$+ C/AB - C/BA - AB/C + B/AC$$

$$[x, (xy - yx)]$$

$$x^2y - xyx - xyx + yx^2$$